

Homework assignment for 02YMECH

Chapter: Dynamics of System of Particles

1. A thin rod of length L lies along the x -axis from $x = 0$ to $x = L$. Its linear mass density is not uniform; instead, it varies as a power of the distance from the left end:

$$\lambda(x) = kx^n, \quad 0 \leq x \leq L,$$

where $k > 0$ is a constant and $n > -1$ is a real parameter.

- (a) Compute the total mass M of the rod in terms of k , n , and L .
 - (b) Find the position of the center of mass x_{CM} as a function of n and L .
 - (c) Suppose that the center of mass is required to be located at $x_{\text{CM}} = \frac{3L}{4}$. Determine the value of n for which this holds. Describe qualitatively how the mass is distributed along the rod for this value of n .
2. Three particles move in the xy -plane. Each particle has mass $m = 1$ kg. Their initial positions at $t = 0$ are

$$\mathbf{r}_1(0) = (1, 0), \quad \mathbf{r}_2(0) = (-1, 0), \quad \mathbf{r}_3(0) = (0, 0),$$

and their velocities for $t \geq 0$ are

$$\mathbf{v}_1(t) = (1, t), \quad \mathbf{v}_2(t) = (-1, t), \quad \mathbf{v}_3(t) = (0, t).$$

The positions of the particles at later times are obtained by integrating the velocities:

$$\mathbf{r}_i(t) = \mathbf{r}_i(0) + \int_0^t \mathbf{v}_i(\tau) d\tau.$$

Using the given time-dependent velocities and positions, determine whether:

- (a) Determine the position of the center of mass as a function of time. Compute its velocity and acceleration. Is the center of mass moving with constant velocity?
 - (b) Determine whether the total linear momentum of the system is conserved.
 - (c) Determine whether the total angular momentum of the system about the origin is conserved.
3. A thin rectangular plate occupies the region $0 \leq x \leq a$, $0 \leq y \leq b$. Its surface mass density varies with x as

$$\sigma(x, y) = \sigma_0 \frac{x}{a},$$

where σ_0 is a constant.

- (a) Find the total mass of the plate.
- (b) Determine the coordinates of the center of mass $(x_{\text{cm}}, y_{\text{cm}})$.